Negative indices

1 Consider the following expressions:

i Identify the base. **ii** Identify the power.

a 10^{-7} **b** 2^{-4} **c** 13^{-10} **d** $(-5)^{-8}$

2 Complete the following tables:

a

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}
32	16					

b

10^5	10^4	10^3	10^2	10^1	10^0	10^{-1}
100 000	10 000					

c

3^3	3^2	3^1	3^0	3^{-1}	3^{-2}	3^{-3}
27	9					

3 Express the following expressions with a positive index:

a 6^{-10} **b** 73^{-14} **c** $(-9)^{-7}$ **d** 9^{-1}

e 17^{-6} **f** 55^{-1} **g** $(-12)^{-8}$ **h** -45^{-5}

i -8^{-11} **j** $(-20)^{-3}$ **k** 7^{-6} **l** $(-5)^{-1}$

4 Express the following expressions with a negative index:

a $\frac{1}{3}$ **b** $\frac{1}{37}$ **c** $\frac{1}{5}$ **d** $\frac{1}{4^7}$

e $\frac{1}{-15^3}$ **f** $\frac{1}{10^5}$ **g** $\frac{1}{(-24)^{10}}$ **h** $\frac{1}{25^3}$

i $\frac{1}{13^{11}}$ **j** $\frac{1}{7^8}$ **k** $\frac{1}{16^{12}}$ **l** $\frac{1}{(-45)^7}$

5 Complete the following statements:

a $\frac{1}{3^2} = 3^{\square}$ **b** $\frac{1}{6^5} = 6^{\square}$ **c** $\frac{1}{27} = 3^{\square}$ **d** $\frac{1}{5^3} = 5^{\square}$

e $\frac{1}{\square} = 7^{\square}$ **f** $\frac{1}{\square} = 4^{\square}$ **g** $\frac{1}{\square} = -9^{\square}$ **h** $\frac{1}{\square} = -2^{\square}$

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6 Simplify the following expressions:

a $5^{11} \div 5^{-3}$ b $7^{-7} \div 7^{5}$ c $7^{-3} \times 7^{-4}$ d $5^{-4} \div 5^{-9}$

e $9^{0} \times 9^{-12}$ f $5^{13} \div 5^{-9}$ g $10^{2} \div 10^{3}$ h $100^{25} \div 100^{26}$

i $\dfrac{2^{3}}{2^{5}}$ j $\dfrac{\left(5^{2}\right)^{9} \times 5^{6}}{5^{40}}$ k $\dfrac{\left(19^{2}\right)^{3}}{19^{-3} \times 19^{-9}}$ l $\dfrac{-\left(8^{5}\right)^{2}}{8^{-2} \times 8^{-12}}$

7 Evaluate:

a $4 \times 3^{-2} + 8^{0}$ b $6 \div 2^{-1} + 4$ c $\left(2 + 3^{-2}\right) \times 2^{-1}$ d $\left(\dfrac{2}{3}\right)^{-3} \div 6^{-1}$

8 Answer the following questions:

a What is 0^{4} equal to?

b Explain why 0^{-4} is undefined.

9 Evaluate the following expressions:

a $6^{7} \times 6^{-7}$ b $4^{8} \times 4^{-6}$ c $3^{-8} \times 3^{11}$ d $5^{5} \times 5^{-7}$

e $4^{-4} \times 2^{-4}$ f $12^{13} \div 12^{7} \div 12^{8}$ g $5^{-2} \times 3^{-2}$ h $(-4)^{3} \times (-4)^{-7}$

i $7^{12} \div 7^{-7} \div 7^{17}$ j $\dfrac{3^{-9} \times 3^{-7}}{3^{-14}}$ k $\dfrac{5^{7} \times 5^{-8}}{5^{-4}}$ l $\dfrac{(-7)^{-11} \times (-7)^{-4}}{(-7)^{-14}}$

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